Backpropagation: worked one-sample derivation

Given $x=1$, target $y=2$, $w=0.5$, $b=0$, $v=1$, $c=0$ and learning rate $\eta=0.1$.

Forward pass

$z=wx+b=0.5$; $h=\tanh(0.5)=0.4621171573$. Prediction $\hat{y}=vh+c=0.4621171573$. Error $e=\hat{y}-y=-1.5378828427$. Loss $J=0.5e^2=1.1825418190$.

Chain rule

$\frac{dJ}{d\hat{y}}=e$. $\frac{dJ}{dv}=e\cdot h=-0.7106820475$. $\frac{dJ}{dc}=e=-1.5378828427$. $\frac{dh}{dz}=1-\tanh^2(z)=1-h^2$. $\frac{dJ}{dw}=e\cdot v\cdot(1-h^2)\cdot x=-1.2094644752$. $\frac{dJ}{db}=e\cdot v\cdot(1-h^2)=-1.2094644752$.

Simultaneous update

Use the gradients above, all evaluated at the original parameters: $w_{new}=w-\eta dw=0.6209464475$; $b_{new}=b-\eta db=0.1209464475$; $v_{new}=v-\eta dv=1.0710682047$; $c_{new}=c-\eta dc=0.1537882843$.

Gradient direction

All four initial gradients are negative. Gradient descent subtracts the gradient, so each parameter increases in this first update. The update moves the prediction toward the target value 2 and reduces the loss. The C++ implementation verifies the initial gradients using central finite differences with epsilon 1e-6.

Study appendix only: manually reproduce this derivation legibly and attach your own scan/photo as required by the assignment. This typed page is not a handwritten submission.